Related Work: Step-Level Feedback Distillation

1 Related Work

1.1 Process supervision and process reward models

A line of work has established that supervising how a model reasons, step by step is more effective than supervising only whether the final answer is correct. Lightman et al (1) compared the two regimes directly and found that process supervision substantially outperforms outcome supervision on the challenge MATH dataset, releasing the large PRM800K corpus of step level human labels to support the finding. Because human step annotation is expensive. Math-Shepherd (2) showed that process supervision can be constructed automatically, whcih is assigning a reward to each solution step without manual labels and used the resulting signal both to rerank candidate solutions and to drive step-level reinforcement learning. These process reward models (PRMs) treat each step's correctness as a **scalar** quantity and deploy the PRM as a frozen scorer or ranker. Our work shares the step level granularity but departs from the scalar view: we treat the teacher step judgment, including its natural language critique, as the object to be distilled rather than a number to be predicted.

1.2 Generative PRMs

A more recent thread makes the verifier itself generative, producing a reasoning trace before it judges. GenPRMs(3) performs explicit chain-of-thought reasoning with code verification before scoring each step, while ThinkPRM(4) fine-tunes a long-COT verifier that articulates a verification rationale for every step and reaches strong ProcessBench performance from only a few thousand process labels. These models are close to uors in spirit they generate critiques rather than emit bare scores - but they are designed as verifiers evaluated at inference time, not as teacher whose score-plus-critique behavior is distilled into a separate, smaller, answer-blind student. Crucially, none of them isolates when priviledge (answer-aware) judgment actually helps, which the question we take up.

1.3 Expert–amateur contrast

Our framing descendds from work that extracts signal from the gap between a stronger and a weaker model. CLEAR (5) contrasts feedback from a larger "expert" and a smaller "amateur" model and applies the refined difference to iteratively improve outputs across reasoning tasks; it is the direct predecessor of the present project. LightReasoner (6) similarly exploits the behavioral divergence between a strong expert and a weak amateur to pinpoint high-value reasoning moments and distill them, which is notably without ground-truth labels, to sharpen reasoning. We retain the ground-truth free constraints at student test time but invert the asymmetry of interest rather than contrasting models of different capacity. We study a teacher with a priviledged view, which has access to reference solution while labeling, distilled into a student that is **answer-blind** by construction. The relevant axis for us is information, not size.

1.4 Our position

Against this backdrop, our contribution is twofold. First, we distill a teacher's step-level score together with its natural language critique into a small, ground-truth-free student that flags which step a solution first goes wrong. A distillation target richer than the scalar reward of standard PRMs and aimed at error localization rather than solution ranking. Second we characterize when privileged supervision is worth anything, and report a privilege x difficulty x richness effect: a teacher's answer-aware judgment helps the student only when the problem is hard and the privilege is rich. On easy problems (GMS8K) capable teacher already self-verifies, and the privilege gap collapses to roughly zero; on hard problems (MATH) a bare final answer still moves nothing, while a full worked reference solution lifts error recall by a clear margin. To our knowledge, this conditional structure that privilege buys signal precisely where self-verification fails, and only when the privileged information is a reasoning trace rather than a number has not been isolated in the PRM or expert-amateur literate.

2 Results

2.1 Setup and metric

We evaluate a teacher's ability to localize the first **erroneous step** in a multi-step solution under three supervision conditions that differ only in the privileged information made available to the teacher at labeling time: (i) **no-GT**, an answer-blind condition in which the teacher sees only the problem and the candidate solution; (ii) **+answer**, in which the teacher is additionally given in the final reference answer; and (iii) **solution**, in which the teacher is given the full worked reference solution. Performance is reported as ProcessBench-style first-error **F1**, toghether with error-detection accuracy. The student remains ground-truth-free at test time throughout; privilege is a property of the teacher's labeling pass, not of the deployed evaluator. Reported deltas are taken against the no-GT condition, which serves as the unprivileged reference point.

2.2 Teacher selection

Before probing the effect of privilege, we fix the teacher by an upper-bound bake-off in which every candidate is granted ground-truth access (the with-GT ceiling, N=30). The official Gemma-class checkpoint attains the strongest first-error F1 (0.909) with perfect format compliance (zero parse failures) and the highest throughput among the candidates; a comparably-sized Qwen variant follows closely (F1 0.873). Notably, a reasoning-tuned candidate collapses under the structured-output protocol, failing to parse on the large majority of instances (parse-failure rate 0.846, F1 0.000). This is a reminder that verbal reasoning fluency does not by itself confer reliable structured step adjudication. We therefore adopt the Gemma checkpoint as the primary teacher and retain the Qwen variant for the cross-family replication reported below

Teacher	**F1**	**correct_acc**	**error_acc**	**parse_fail**
Gemma-4-26b (primary)	**0.909**	1.000	0.833	0.000
Qwen-27B	0.873	0.917	0.833	0.000
Qwen-35B-A3B	0.839	1.000	0.722	0.019
Qwen-4B-Thinking	0.000	1.000	0.000	0.846

2.3 Privilege helps only where self-verification fails

Our central finding is that privileged supervision is not uniformly beneficial: its value is contingent on problem difficulty. On GSM8K, privilege confers no advantage. Relative to the answer-blind baseline, neither the bare answer nor the full solution improves first-error detection; if anything, the additional context marginally perturbs an already-saturated judgment (solution-condition $\Delta \approx -0.03F1$, $\mathrm{N} = 50$). In this regime the teacher is, in effect, its own oracle, and external privilege is redundant.

MATH (Gemma, $N = 150$)	**F1**	**err_acc**
no-GT	0.889	0.800
+answer	0.856	0.800
+solution	0.837	0.720
Δ (solution $-$ no-GT)	-0.052	-0.080

The picture inverts on MATH, where problems are hard enough that unaided self-verification degrades. Here privileged access to the full worked solution produces a clear and consistent gain. On the primary run (N=150), the full solution lifts first-error F1 from 0.716 to 0.786 (+0.07) and error recall from 0.577 to 0.654 (+0.08) — the teacher recovers roughly one in seven of the errors it had previously missed. The smaller preliminary run (N=50) points the same direction with a larger nominal gap (+0.10 F1, +0.13 err$_{a}cc); wetreattheN = 150figuresastheconservative, headlineresult.$

MATH (Gemma, $N = 150$)	**F1**	**err_acc**
no-GT	0.716	0.577
+answer	0.716	0.577
+solution	0.786	0.654
Δ (solution $-$ no-GT)	$+0.070$	$+0.077$

The contrast between these two panels is the paper's load-bearing result: **privilege buys signal precisely where the teacher cannot already self-verify**. On easy problems the privilege gap collapses to zero; on hard problems it opens. Difficulty is the moderator, not an incidental co-variate.

2.4 Richness, not mere access, is what matters

The MATH table also isolates a second, sharper effect. The **+answer** condition, in which the teacher is handed the correct final answer but no reasoning, is statistically indistinguishable from teh answer-blind baseline: at N=150 the two conditions yield identical F1 and error recall (0.716 / 0.577), and inspection confirms that the bare answer flips zero predictions. (We verified the labeling pipeline is correctly wired; the null effect is behavioral, not an artifact.). Only the full worked solution moves the metric. The operative variable is therefore not whether the teacher is privileged but whether the privilege is rich: a reference reasoning trace supplies the step-by-step scaffold needed to localize an error, wheras a final number, which the teacher cannot back-propagate into the faulty step, supplies nothing it can act on. We summarize the joint effect as a **privilege x difficulty x richness** interaction: privileged supervision helps a step-error-detecting teacher only when the problem is hard and the priviledged information is a worked solution rather than a bare answer.

2.5 Cross-family replication

To establish that this interaction reflects a property of step-level feedback under difficulty rather than an idiosyncrasy of the Gemma family, we replicate the MATH probe (N = 150) with the Qwen-27B teacher. The pattern holds: the full worked solution lifts first-error F1 from 0.653 to 0.735 (+0.08) and error detection from 0.562 to 0.667, while a bare answer yield only a marginal change, which is consistent with the Gemma result. That the difficulty- and richness-conditioned gain reproduces across two independent model families substantially weakens any single-model-artifact explanation and supports reading the effect as a general characteristic of privileged step-level supervision.

MATH (Qwen-27B, N = 150)	**F1**	**err_acc**
no-GT	0.653	0.526
+answer	0.690	0.564
+solution	0.735	0.667
Δ (solution − no-GT)	+0.082	+0.141

References

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